The documentation and process conversion measures necessary to comply with this revision shall be completed by 24 May 1995

INCH-POUND

MIL-S-19500/439D
24 February 1995
SUPERSEDING
MIL-S-19500/439C
28 May 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN SILICON, HIGH-POWER
TYPES 2N5038 AND 2N5039
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon , high power transistors for use in high-speed power switching applications. Four levels of product assurance are provided for each device type as specified herein and in MIL-S-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See 3.3, figure 1 (TO-3), and figure 2 (JANHC and JANKC).

1.3 Maximum ratings.

Types	P_T 1/ at T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{STG} and T_{OP}
	W	V dc	V dc	V dc	A dc	A dc	°C
2N5038	140	150	90	7.0	5	20	-65 to +200
2N5039	140	125	75	7.0	5	20	-65 to +200

1/ Derate linearly 800 mW/°C for T_C > 25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	h_{FE3} 1/ V_{CE} = 5.0 V dc, I_C = 12 A dc (2N5038), I_C = 10 A dc (2N5039)	$\|h_{fe}\|$ V_{CE} = 10 V dc, I_C = 2.0 A dc, f = 5 MHz	$V_{CE(sat)1}$ 1/ I_C = 12 A dc, I_B = 1.2 A dc (2N5038), I_C = 10 A dc, I_B = 1.0 A dc (2N5039)	C_{obo} V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 mHz	Pulse Response		$R_{\theta JC}$
					t_{on}	t_{off}	
			V dc	pf	µs	µs	°C/W
Minimum	20	12					1.25
Maximum		48	1.0	500	0.5	2.0	

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

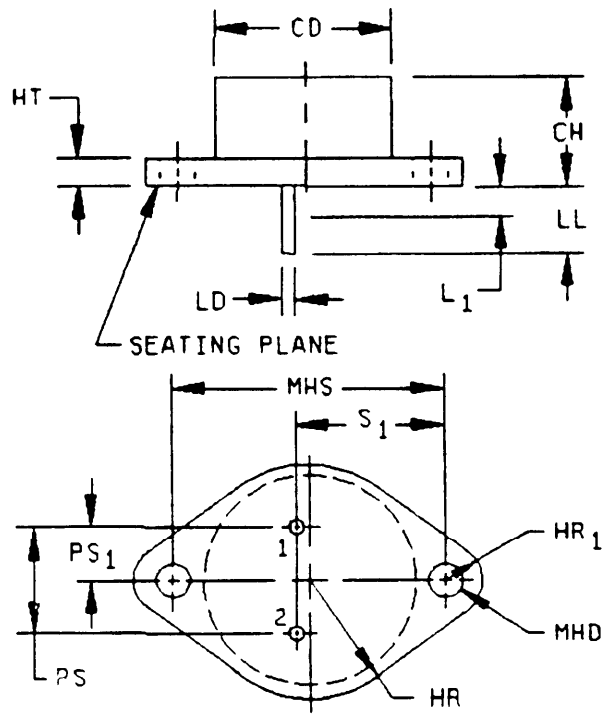
AMSC N/A

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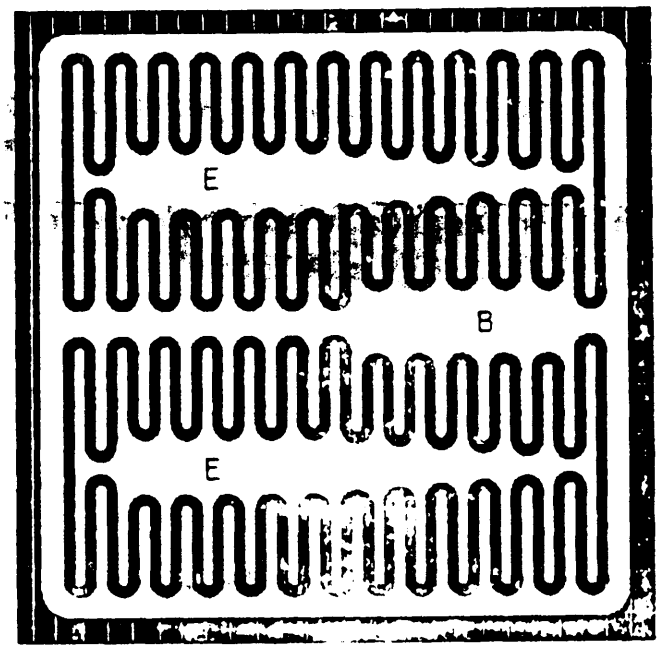
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.875		22.23	
CH	.250	.380	6.35	9.65	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.050	.135	1.27	3.43	
LD	.038	.043	.97	1.09	
LL	.312	.500	7.92	12.70	
L_1		.050		1.27	
MHD	.151	.161	3.84	4.09	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	1
PS_1	.205	.225	5.21	5.72	1
S_1	.655	.675	16.64	17.15	



NOTES:
1. These dimensions should be measured at points .0.5 (1.27 mm) .055 (1.40 mm) below seating plane. When a gauge is not used, measurement will be made at the seating plane.
2. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5 Collector shall be electrically connected to the case.
6 Lead designation: lead 1 = emitter, lead 2 = base, lead 3 (case)= collector

FIGURE 1. Physical dimensions (similar to TO-3).

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NOTES:

1.	Chip size:	180 x 180 mils ± 2 mils
2.	Chip thickness:	5 to 9 mils
3.	Top metal:	Aluminum 40,000 Å min, 50,000 Å nominal
4.	Back metal:	Gold 2,500 Å min, 3,000 Å nominal
5.	Backside:	Collector
6.	Bonding pad:	B = 17 x 60 mils ± 2 mils, E = 17 x 70 mils ± 2 mils

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction and physical dimensions shall be as specified in figures 1 and 2 (TO-3 and JANHC/JANKC) herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2.1 JANHC and JANKC devices. JANHC and JANKC devices shall be qualified in accordance with MIL-S-19500.

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4.3 Screening. Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CEX1}, h_{FE2}, and h_{FE3}	I_{CEX1}
11	I_{CEX1}, h_{FE2}, and h_{FE3}; ΔI_{CEX1} = 100% of initial value or 1 µA dc, whichever is greater; Δh_{FE3} = ±20 of the initial value.	I_{CEX1}, h_{FE2}, and h_{FE3}; ΔI_{CEX1} = 100% of initial value or 1 µA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100% of initial value or 1 µA dc, whichever is greater; Δh_{FE3} = ±20% of the initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or 1 µA dc, whichever is greater; Δh_{FE3} = ±20% of the initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +187.5°C ±12.5°C; V_{CB} = 60 V dc; $T_A \leq 100°C$.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV), of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Test condition A
B4	1037	2,000 cycles; V_{CB} = 10 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ ≥ 1 minute.
B5	1027	V_{CE} ≥ 20 V dc; P_T = 80 W at T_C = 100°C or adjusted as required by the chosen T_C to give an average lot T_J = 225°C.
B7	3053	Load condition C; (unclamped inductive load) (see figure 7); T_C = 25°C; duty cycle ≤ 10%; Rs = 0.1Ω; $t_r = t_p$ ≤ 500 ns. TEST 1: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 2Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 20 A; L = 70 µH, 0.1Ω. TEST 2: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 40Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 4.5 A; L = 500 µH, 0.1Ω.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Test condition A
B3	1037	2,000 cycles; V_{CB} = 10 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ ≥ 1 minute.
B7	3053	Load condition C; (unclamped inductive load) (see figure 7); T_C = 25°C; duty cycle ≤ 10%; Rs = 0.1Ω; $t_r = t_p$ ≤ 500 ns. TEST 1: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 2Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 20 A; L = 70 µH, 0.1Ω. TEST 2: t_p = 5 ms (vary to obtain I_C); R_{BB1} = 40Ω; V_{BB1} = 10 V dc; R_{BB2} = 20Ω; V_{BB2} = 4 V dc; V_{CC} = 10 V dc; I_C = 4.5 A; L = 500 µH, 0.1Ω.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	1056	Test condition B
C2	2036	Test condition A; weight = 10 pounds; time 15 s.
C6	1027	6,000 cycles; V_{CE} = 10 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ ≥ 1 minute.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 200 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N5038				90		V dc
2N5039				75		V dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		1	µA dc
2N5038		V_{CE} = 70 V dc				
2N5039		V_{CE} = 55 V dc				
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		1	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc	I_{CEX1}		5	µA dc
2N5038		V_{CE} = 100 V dc				
2N5039		V_{CE} = 85 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO}		1	µA dc
2N5038		V_{CE} = 150 V dc				
2N5039		V_{CE} = 125 V dc				
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 25 ma dc	$V_{(BR)EBO}$	7.0		V dc
Base to emitter voltage (non-saturated)	3066	Test condition B, V_{CE} = 5 V dc; pulsed (see 4.5.1)	V_{BE}		1.8	V dc
2N5038		I_C = 12 A dc				
2N5039		I_C = 10 A dc				
Base to emitter voltage (saturated)	3066	Test condition A, I_C = 20 A dc; I_B = 5 a dc	$V_{BE(sat)}$		3.3	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
2N5038		I_C = 12 A dc; I_B = 1.2 A dc				
2N5039		I_C = 10 A dc; I_B = 1.0 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 -Continued						
Collector to emitter voltage (saturated)	3071	I_C = 20 A dc; I_B = 5.0 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE1}			
2N5038				50		
2N5039				30		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 2.0 A dc; pulsed (see 4.5.1)	h_{FE2}			
2N5038				50	200	
2N5039				30	150	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; pulsed (see 4.5.1)	h_{FE3}	15		
2N5038		I_C = 12 A dc				
2N5039		I_C = 10 A dc				
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX2}		100	µA dc
2N5038		V_{CE} = 100 V dc				
2N5039		V_{CE} = 85 V dc				
Low-temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; pulsed (see 4.5.1)	h_{FE4}	10		
2N5038		I_C = 12 A dc				
2N5039		I_C = 10 A dc				
Subgroup 4						
Magnitude of small signal short-forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 2 A dc; f = 5 MHz	$\|h_{FE}\|$	12	48	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		500	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Pulse response:	3251	Test condition A except test circuit and pulse requirement (see figure 3)				
Turn-on time		V_{CC} = 30 ±2 V dc	t_{on}		0.5	µs
2N5038		I_C = 12 A dc; I_{B1} = 1.2 A dc				
2N5039		I_C = 10 A dc; I_{B1} = 1.0 A dc				
Turn-off time		V_{CC} = 30 ±2 V dc	t_{off}		2.0	µs
2N5038		I_C = 12 A dc $I_{B1} = -I_{B2}$ = 1.2 A dc				
2N5039		I_C = 10 A dc $I_{B1} = -I_{B2}$ = 1.0 A dc				
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25°C; t = 1 s; 1 cycle, (see figure 4)				
Test 1		I_C = 5 A dc; V_{CE} = 28 V dc				
Test 2		I_C = 0.9 A dc; V_{CE} = 45 V dc				
Test 3		I_C = 20 A dc; V_{CE} = 7.0 V dc				
Test 4						
2N5038 (only)		I_C = 0.23 A dc; V_{CE} = 90 V dc				
Test 5						
2N5039 (only)		I_C = 0.32 A dc; V_{CE} = 75 V dc				
Safe operating area (clamped switching)		T_C = 25°C; I_C = 20 A dc; (see figures 5 and 6)				
2N5038		clamp voltage = 90 V dc				
2N5039		clamp voltage = 75 V dc (Device fails if clamp voltage is not reached)				
Electrical measurements		See table II, steps 1 and 3				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to emitter cutoff current 2N5038 2N5039	3041	Bias condition A; V_{BE} = -1.5 V dc V_{CE} = 100 V dc V_{CE} = 85 V dc	I_{CEX1}		5	µA dc
2	Collector to emitter cutoff current 2N5038 2N5039	3041	Bias condition A; V_{BE} = -1.5 V dc V_{CE} = 100 V dc V_{CE} = 85 V dc	I_{CEX2}		10	µA dc
3	Forward-current transfer ratio 2N5038 2N5039	3076	V_{CE} = 5.0 V dc; pulsed (see 4.5.1) I_C = 12 A dc I_C = 10 A dc	h_{FE3}	15		
4	Base emitter voltage (saturated)	3066	Test condition A; I_C = 20 A dc; I_B = 5 A dc	$V_{BE(sat)}$		3.3	V dc
5	Collector to emitter voltage (saturated) 2N5038 2N5039	3071	pulsed (see 4.5.1) I_C = 12 A dc I_B = 1.2 A dc I_C = 10 A dc I_B = 1.0 A dc	$V_{CE(sat)1}$		1.0	V dc
6	Collector to emitter voltage (saturated)	3071	See step 5	$\Delta V_{CE(sat)1}$	50 mV dc change from initial value.		
7	Collector to emitter cutoff current	3041	See step 1	ΔI_{CEX1}	100% of initial value or 1µA dc, whichever is greater.		
8	Forward-current transfer ratio	3076	See step 3	Δh_{FE3}	±25% change from initial value.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, and 5.
b. Subgroup 4, see table II herein, steps 1, 3, 4, 5, and 6.
c. Subgroup 5, see table II herein, steps 1, 3, 4, 5, 6, 7, and 8.
d. Subgroup 7, see table II herein, steps 1 and 3.

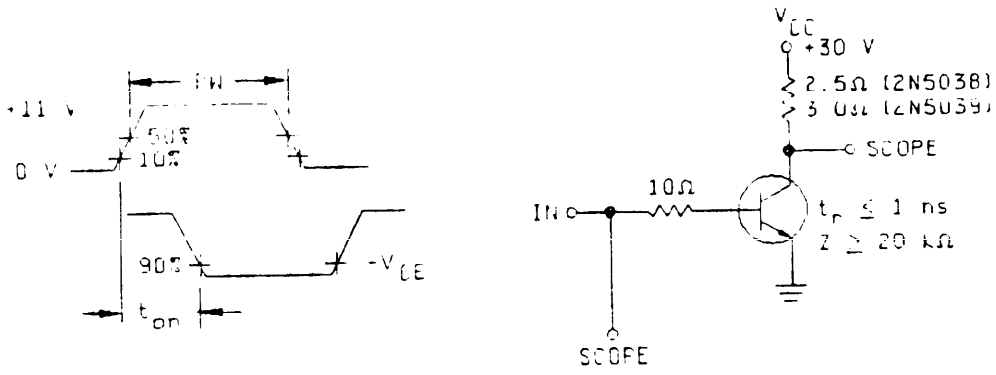
2/ The electrical measurements for table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 are as follows:

a. Subgroups 2 and 7, see table II herein, steps 1 and 3.
b. Subgroups 3 and 6, see table II herein, steps 2, 3, and 8.

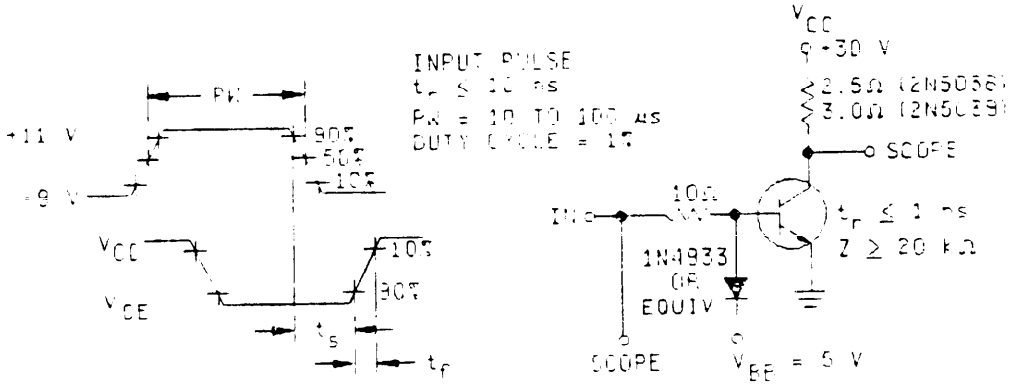
3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1, 3, 4, and 5 (JANS) and steps 1 and 3 (JAN, JANTX, and JANTXV).
b. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, 7, and 8 (JANS) and steps 2, 3, and 8 (JAN, JANTX, and JANTXV).

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TURN-ON t_{on} TIME TEST CIRCUIT



TURN-OFF t_{off} TIME TEST CIRCUIT

FIGURE 3. Switching time test circuits.

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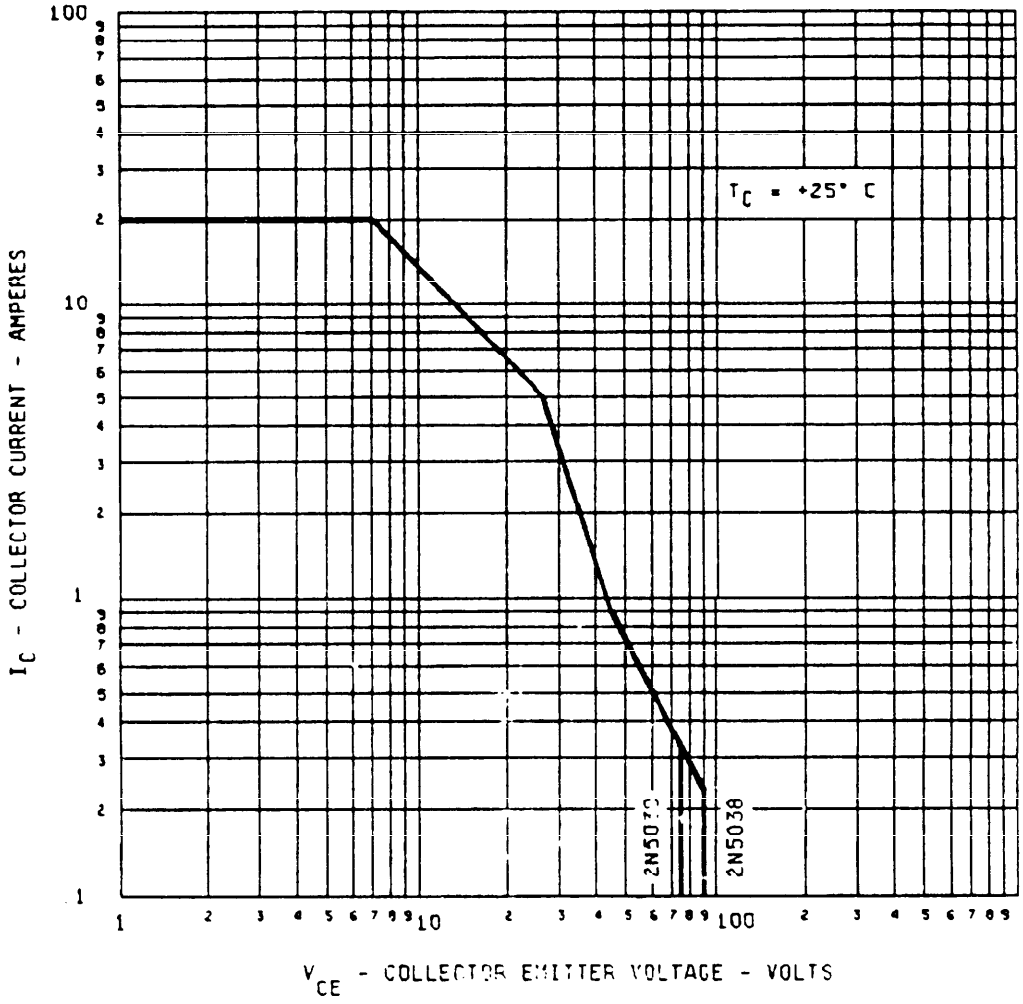
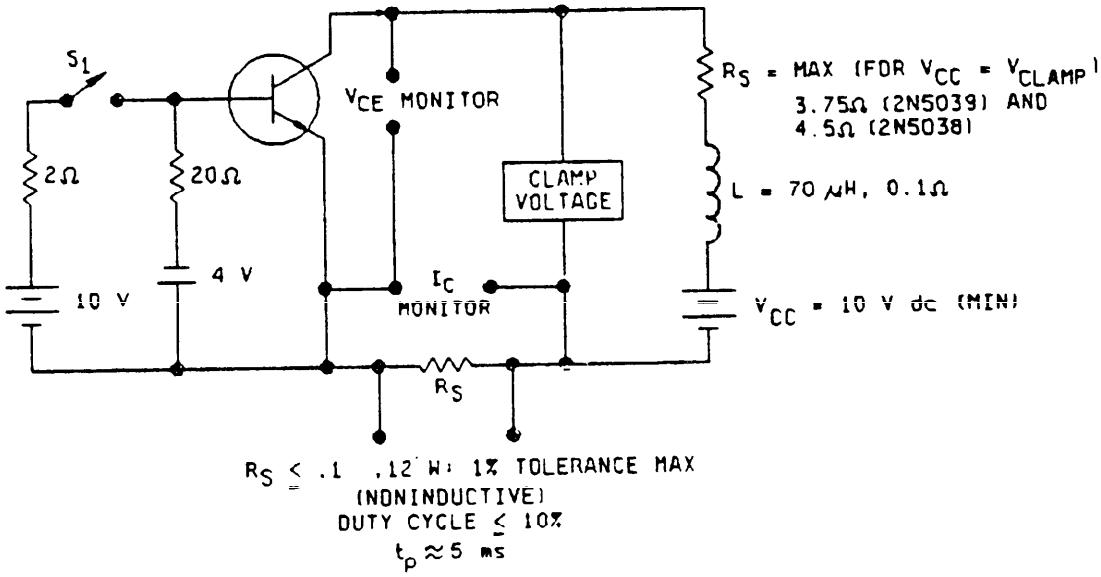


FIGURE 4. Maximum safe operating area graph (continuous dc).

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NOTES:
1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached.
3. Perform specified end-point tests.

FIGURE 5. Clamped inductive sweep test circuit.

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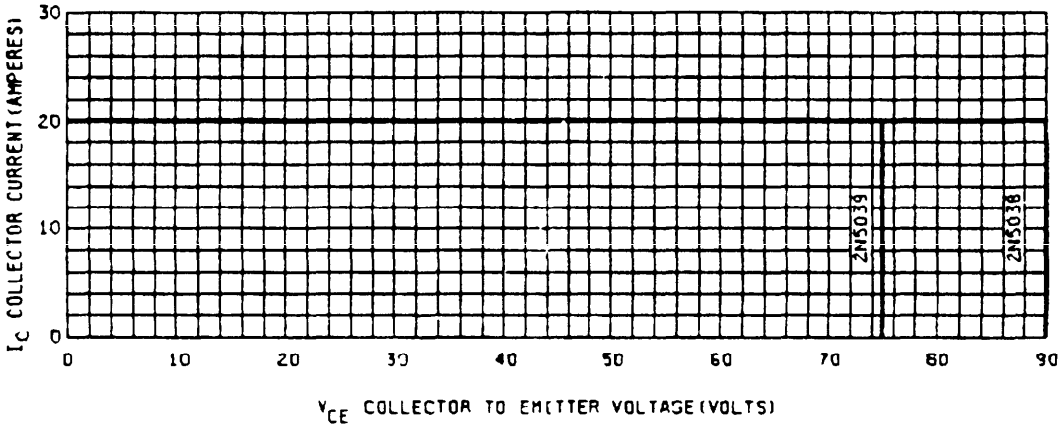


FIGURE 6. Safe operating area for switching between saturation and cutoff - clamped inductive load.

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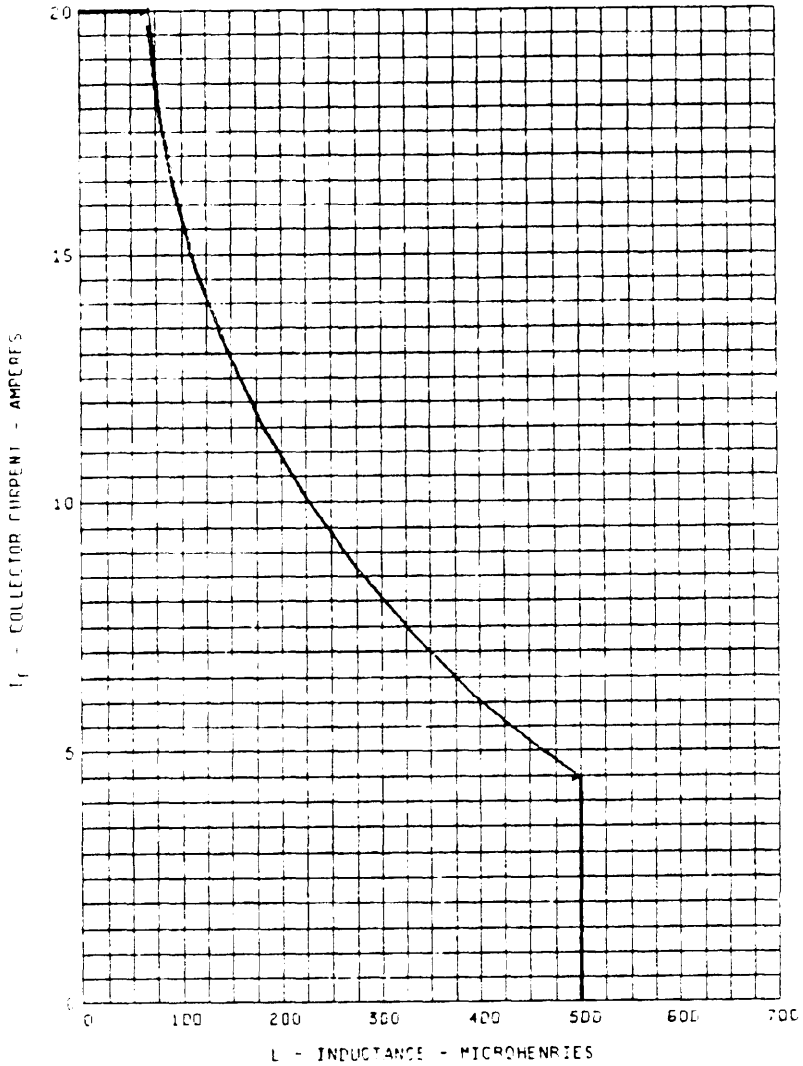


FIGURE 7. Safe operating area for switching between saturation and cutoff - unclamped inductive load.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Minimum collector current magnitude during power applications shall be 2 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

e. Mounting arrangements shall be with heat sink to case.

f. Maximum limit $R_{\Theta JC}$ shall be 1.25°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designation.

6.3 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5038) will be identified on the QPL.

JANHC and JANKC ordering information		
	Manufacturer	
PIN	33173	
2N5038	JANHCA2N5038	
2N5039	JANHCA2N5039	
2N5038	JANKCA2N5038	
2N5039	JANKCA2N5039	

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC

Review activities:
Army - MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 80, 85, 99
DLA - ES

Preparing activity:
NASA -NA

Agent:
DLA - ES

(Project 5961-1692)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/439D	2. DOCUMENT DATE (YYMMDD) 95/02/24

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN SILICON, HIGH-POWER TYPES 2N5038 AND 2N5039 JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Gregory Rose	b. TELEPHONE (Include Area Code) (1) Commercial 301-731-8694	(2) AUTOVON
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO) NASA Goddard Space Flight Center, Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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